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The Sommerfeld Theory of Metals

Fermi-Dirac Distribution

Free Electrons

Density of Allowed Wave Vectors

Fermi Momentum, Energy, and Temperature

Ground-State Energy and Bulk Modulus

Thermal Properties of a Free Electron Gas

Sommerfeld Theory of Conduction

Wiedemann-Franz Law

In Drude's time, and for many years thereafter, it seemed reasonable to assume that the electronic velocity distribution, like that of an ordinary classical gas of density $n = N/V$, was given in equilibrium at temperature T by the Maxwell-Boltzmann distribution. This gives the number of electrons per unit volume with velocities in the range¹ $d\mathbf{v}$ about $\mathbf{v}$ as $f_B(\mathbf{v})d\mathbf{v}$, where

$$f_B(\mathbf{v}) = n \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-mv^2/2k_B T}. \quad (2.1)$$

We saw in Chapter 1 that in conjunction with the Drude model this leads to good order of magnitude agreement with the Wiedemann-Franz law, but also predicts a contribution to the specific heat of a metal of $\frac{3}{2}k_B$ per electron that was not observed.²

This paradox cast a shadow over the Drude model for a quarter of a century, which was only removed by the advent of the quantum theory and the recognition that for electrons³ the Pauli exclusion principle requires the replacement of the Maxwell-Boltzmann distribution (2.1) with the Fermi-Dirac distribution:

$$f(\mathbf{v}) = \frac{(m/\hbar)^3}{4\pi^3} \frac{1}{\exp[(\frac{1}{2}mv^2 - k_B T_0)/k_B T] + 1}. \quad (2.2)$$

Here $\hbar$ is Planck's constant divided by 2π , and T_0 is a temperature that is determined by the normalization condition⁴

$$n = \int d\mathbf{v} f(\mathbf{v}), \quad (2.3)$$

and is typically tens of thousands of degrees. At temperatures of interest (that is, less than 10^3 K) the Maxwell-Boltzmann and Fermi-Dirac distributions are spectacularly different at metallic electronic densities (Figure 2.1).

In this chapter we shall describe the theory underlying the Fermi-Dirac distribution (2.2) and survey the consequences of Fermi-Dirac statistics for the metallic electron gas.

Shortly after the discovery that the Pauli exclusion principle was needed to account for the bound electronic states of atoms, Sommerfeld applied the same principle to the free electron gas of metals, and thereby resolved the most flagrant thermal anomalies of the early Drude model. In most applications Sommerfeld's model is nothing more than Drude's classical electron gas with the *single* modification that the electronic velocity distribution is taken to be the quantum Fermi-Dirac distribution

¹ We use standard vector notation. Thus by v we mean the magnitude of the vector $\mathbf{v}$; a velocity is in the range $d\mathbf{v}$ about $\mathbf{v}$ if its i th component lies between v_i and $v_i + dv_i$, for $i = x, y, z$; we also use $d\mathbf{v}$ to denote the volume of the region of velocity space in the range $d\mathbf{v}$ about $\mathbf{v}$: $d\mathbf{v} = dv_x dv_y dv_z$ (thereby following the practice common among physicists of failing to distinguish notationally between a region and its volume, the significance of the symbol being clear from context).

² Because, as we shall see, the actual electronic contribution is about 100 times smaller at room temperature, becoming smaller still as the temperature drops.

³ And any other particles obeying Fermi-Dirac statistics.

⁴ Note that the constants in the Maxwell-Boltzmann distribution (2.1) have already been chosen so that (2.3) is satisfied. Equation (2.2) is derived below; see Eq. (2.89). In Problem 3d the prefactor appearing in Eq. (2.2) is cast in a form that facilitates direct comparison with Eq. (2.1).

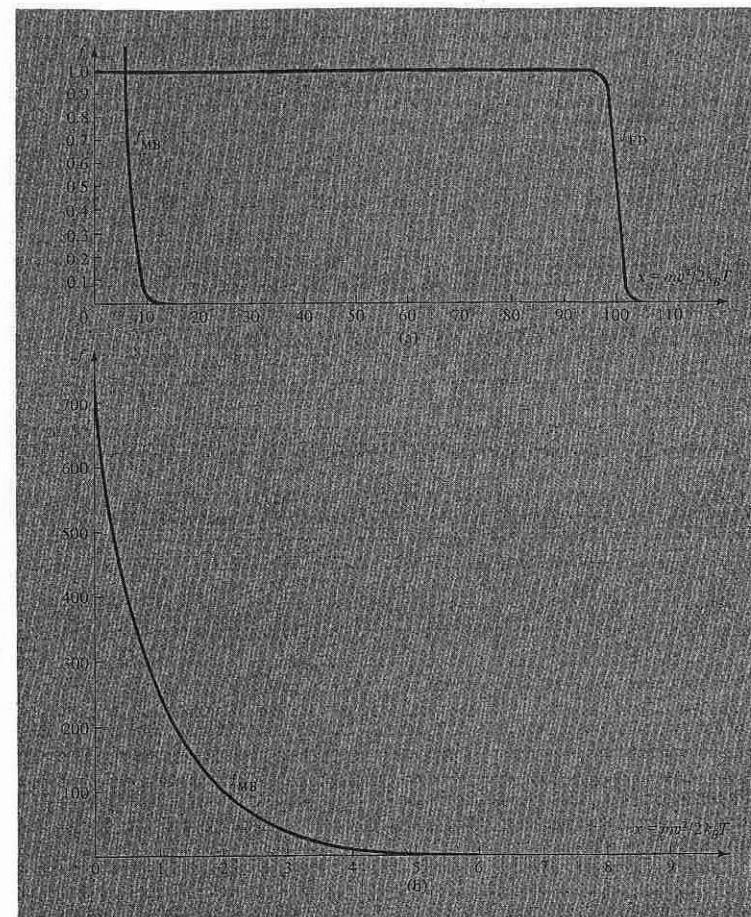


Figure 2.1

(a) The Maxwell-Boltzmann and Fermi-Dirac distributions for typical metallic densities at room temperature. (Both curves are for the density given by $T = 0.01T_0$.) The scale is the same for both distributions, and has been normalized so that the Fermi-Dirac distribution approaches 1 at low energies. Below room temperature the differences between the two distributions are even more marked. (b) A view of that part of (a) between $x = 0$ and $x = 10$. The x -axis has been stretched by about a factor of 10, and the f -axis has been compressed by about 500 to get all of the Maxwell-Boltzmann distribution in the figure. On this scale the graph of the Fermi-Dirac distribution is indistinguishable from the x -axis.

rather than the classical Maxwell-Boltzmann distribution. To justify both the use of the Fermi-Dirac distribution and its bold grafting onto an otherwise classical theory, we must examine the quantum theory of the electron gas.⁵

For simplicity we shall examine the ground state (i.e., $T = 0$) of the electron gas before studying it at nonzero temperatures. As it turns out, the properties of the ground state are of considerable interest in themselves: we shall find that room temperature, for the electron gas at metallic densities, is a very low temperature indeed, for many purposes indistinguishable from $T = 0$. Thus many (though not all) of the electronic properties of a metal hardly differ from their values at $T = 0$, even at room temperature.

GROUND-STATE PROPERTIES OF THE ELECTRON GAS

We must calculate the ground-state properties of N electrons confined to a volume V . Because the electrons do not interact with one another (independent electron approximation) we can find the ground state of the N electron system by first finding the energy levels of a single electron in the volume V , and then filling these levels up in a manner consistent with the Pauli exclusion principle, which permits at most one electron to occupy any single electron level.⁶

A single electron can be described by a wave function $\psi(\mathbf{r})$ and the specification of which of two possible orientations its spin possesses. If the electron has no interactions, the one electron wave function associated with a level of energy ε satisfies the time-independent Schrödinger equation⁷:

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(\mathbf{r}) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) = \varepsilon \psi(\mathbf{r}). \quad (2.4)$$

We shall represent the confinement of the electron (by the attraction of the ions) to the volume V by a boundary condition on Eq. (2.4). The choice of boundary condition, whenever one is dealing with problems that are not explicitly concerned with effects of the metallic surface, is to a considerable degree at one's disposal and can be determined by mathematical convenience, for if the metal is sufficiently large we should expect its *bulk* properties to be unaffected by the detailed configuration of its surface.⁸ In this spirit we first select the shape of the metal to suit our analytic convenience. The time-honored choice is a cube⁹ of side $L = V^{1/3}$.

Next we must append a boundary condition to the Schrödinger equation (2.4),

⁵ Throughout this chapter we shall take "electron gas" to mean a gas of free and independent electrons (see page 4) unless we are explicitly considering corrections due to electron-electron or electron-ion interactions.

⁶ Note that here and later we shall reserve the term "state" for the state of the N -electron system, and the term "level" for a one-electron state.

⁷ We also make the free electron approximation, so that no potential energy term appears in the Schrödinger equation.

⁸ This is the approach that is almost universally followed in theories of macroscopic matter. Rigorous proofs that bulk properties are independent of the boundary conditions can now be constructed in a variety of contexts. The work most pertinent to solid state physics is by J. L. Lebowitz and E. H. Lieb, *Phys. Rev. Lett.* **22**, 631 (1969).

⁹ We shall subsequently find it far more convenient to take not a cube but a parallelepiped with

reflecting the fact that the electron is confined to this cube. We also make this choice in the belief that it will not affect calculated bulk properties. One possibility is to require the wave function $\psi(\mathbf{r})$ to vanish whenever $\mathbf{r}$ is on the surface of the cube. This, however, is often unsatisfactory, for it leads to standing-wave solutions of (2.4), while the transport of charge and energy by the electrons is far more conveniently discussed in terms of running waves. A more satisfactory choice is to emphasize the inconsequence of the surface by disposing of it altogether. We can do this by imagining each face of the cube to be joined to the face opposite it, so that an electron coming to the surface is not reflected back in, but leaves the metal, simultaneously reentering at a corresponding point on the opposite surface. Thus, if our metal were one-dimensional, we would simply replace the line from 0 to L to which the electrons were confined, by a circle of circumference L . In three dimensions the geometrical embodiment of the boundary condition, in which the three pairs of opposite faces on the cube are joined, becomes topologically impossible to construct in three-dimensional space. Nevertheless, the analytic form of the boundary condition is easily generalized. In one dimension the circular model of a metal results in the boundary condition $\psi(x + L) = \psi(x)$, and the generalization to a three-dimensional cube is evidently

$$\begin{aligned} \psi(x, y, z + L) &= \psi(x, y, z), \\ \psi(x, y + L, z) &= \psi(x, y, z), \\ \psi(x + L, y, z) &= \psi(x, y, z). \end{aligned} \quad (2.5)$$

Equation (2.5) is known as the Born-von Karman (or periodic) boundary condition. We shall encounter it often (sometimes in a slightly generalized⁹ form).

We now solve (2.4) subject to the boundary condition (2.5). One can verify by differentiation that a solution, neglecting the boundary condition, is

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (2.6)$$

with energy

$$\varepsilon(\mathbf{k}) = \frac{\hbar^2 k^2}{2m}, \quad (2.7)$$

where $\mathbf{k}$ is any position independent vector. We have picked the normalization constant in (2.6) so that the probability of finding the electron *somewhere* in the whole volume V is unity:

$$1 = \int d\mathbf{r} |\psi(\mathbf{r})|^2. \quad (2.8)$$

To see the significance of the vector $\mathbf{k}$, note that the level $\psi_{\mathbf{k}}(\mathbf{r})$ is an eigenstate of the momentum operator,

$$\mathbf{p} = \frac{\hbar}{i} \frac{\partial}{\partial \mathbf{r}} = \frac{\hbar}{i} \nabla, \quad \left(p_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \text{ etc. } \right), \quad (2.9)$$

edges not necessarily equal or perpendicular. For the moment we use a cube to avoid minor geometrical complexities, but it is a useful exercise to verify that all the results of this section remain valid for the parallelepiped.

with eigenvalue $\mathbf{p} = \hbar \mathbf{k}$, for

$$\frac{\hbar}{i} \frac{\partial}{\partial \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} = \hbar \mathbf{k} e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (2.10)$$

Since a particle in an eigenstate of an operator has a definite value of the corresponding observable given by the eigenvalue, an electron in the level $\psi_{\mathbf{k}}(\mathbf{r})$ has a definite momentum proportional to $\mathbf{k}$:

$$\mathbf{p} = \hbar \mathbf{k}, \quad (2.11)$$

and a velocity $\mathbf{v} = \mathbf{p}/m$ of

$$\mathbf{v} = \frac{\hbar \mathbf{k}}{m}. \quad (2.12)$$

In view of this the energy (2.7) can be written in the familiar classical form,

$$\varepsilon = \frac{p^2}{2m} = \frac{1}{2} m v^2. \quad (2.13)$$

We can also interpret $\mathbf{k}$ as a wave vector. The plane wave $e^{i\mathbf{k} \cdot \mathbf{r}}$ is constant in any plane perpendicular to $\mathbf{k}$ (since such planes are defined by the equation $\mathbf{k} \cdot \mathbf{r} = \text{constant}$) and it is periodic along lines parallel to $\mathbf{k}$, with wavelength

$$\lambda = \frac{2\pi}{k}, \quad (2.14)$$

known as the de Broglie wavelength.

We now invoke the boundary condition (2.5). This permits only certain discrete values of $\mathbf{k}$, since (2.5) will be satisfied by the general wave function (2.6) only if

$$e^{ik_x L} = e^{ik_y L} = e^{ik_z L} = 1. \quad (2.15)$$

Since $e^z = 1$ only if $z = 2\pi i n$, where n is an integer,¹⁰ the components of the wave vector $\mathbf{k}$ must be of the form:

$$k_x = \frac{2\pi n_x}{L}, \quad k_y = \frac{2\pi n_y}{L}, \quad k_z = \frac{2\pi n_z}{L}, \quad n_x, n_y, n_z \text{ integers}. \quad (2.16)$$

Thus in a three-dimensional space with Cartesian axes k_x , k_y , and k_z (known as k -space) the allowed wave vectors are those whose coordinates along the three axes are given by integral multiples of $2\pi/L$. This is illustrated (in two dimensions) in Figure 2.2.

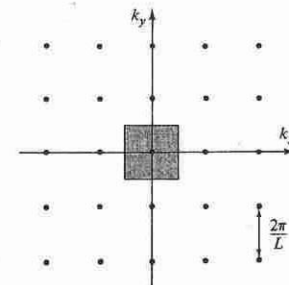
Generally the only practical use one makes of the quantization condition (2.16) is this: One often needs to know how many allowed values of $\mathbf{k}$ are contained in a region of k -space that is enormous on the scale of $2\pi/L$, and that therefore contains a vast number of allowed points. If the region is very large,¹¹ then to an excellent approximation the number of allowed points is just the volume of k -space contained within the region, divided by the volume of k -space per point in the network of

¹⁰ We shall always use the word "integer" to mean the negative integers and zero, as well as the positive integers.

¹¹ And not too irregularly shaped; only a negligible fraction of the points should be within $O(2\pi/L)$ of the surface.

Figure 2.2

Points in a two-dimensional k -space of the form $k_x = 2\pi n_x/L$, $k_y = 2\pi n_y/L$. Note that the area per point is just $(2\pi/L)^2$. In d dimensions the volume per point is $(2\pi/L)^d$.



allowed values of $\mathbf{k}$. That latter volume (see Figure 2.2) is just $(2\pi/L)^3$. We therefore conclude that a region of k -space of volume Ω will contain

$$\frac{\Omega}{(2\pi/L)^3} = \frac{\Omega V}{8\pi^3} \quad (2.17)$$

allowed values of $\mathbf{k}$, or, equivalently, that the number of allowed k -values per unit volume of k -space (also known as the k -space density of levels) is just

$$\frac{V}{8\pi^3}. \quad (2.18)$$

In practice we shall deal with k -space regions so large ($\sim 10^{22}$ points) and so regular (typically spheres) that to all intents and purposes (2.17) and (2.18) can be regarded as exact. We shall begin to apply these important counting formulas shortly.

Because we assume the electrons are noninteracting we can build up the N -electron ground state by placing electrons into the allowed one-electron levels we have just found. The Pauli exclusion principle plays a vital role in this construction (as it does in building up the states of many electron atoms): we may place at most one electron in each single electron level. The one-electron levels are specified by the wave vectors $\mathbf{k}$ and by the projection of the electron's spin along an arbitrary axis, which can take either of the two values $\hbar/2$ or $-\hbar/2$. Therefore associated with each allowed wave vector $\mathbf{k}$ are two electronic levels, one for each direction of the electron's spin.

Thus in building up the N -electron ground state we begin by placing two electrons in the one-electron level $\mathbf{k} = 0$, which has the lowest possible one-electron energy $\varepsilon = 0$. We then continue to add electrons, successively filling the one-electron levels of lowest energy that are not already occupied. Since the energy of a one-electron level is directly proportional to the square of its wave vector (see (2.7)), when N is enormous the occupied region will be indistinguishable from a sphere.¹² The radius of this sphere is called k_F (F for Fermi), and its volume Ω is $4\pi k_F^3/3$. According to (2.17) the number of allowed values of $\mathbf{k}$ within the sphere is

$$\left(\frac{4\pi k_F^3}{3}\right) \left(\frac{V}{8\pi^3}\right) = \frac{k_F^3}{6\pi^2} V. \quad (2.19)$$

¹² If it were not spherical it would not be the ground state, for we could then construct a state of lower energy by moving the electrons in levels farthest away from $\mathbf{k} = 0$ into unoccupied levels closer to the origin.

Since each allowed k -value leads to two one-electron levels (one for each spin value), in order to accommodate N electrons we must have

$$N = 2 \cdot \frac{k_F^3}{6\pi^2} V = \frac{k_F^3}{3\pi^2} V. \quad (2.20)$$

Thus if we have N electrons in a volume V (i.e., an electronic density $n = N/V$), then the ground state of the N -electron system is formed by occupying all single-particle levels with k less than k_F , and leaving all those with k greater than k_F unoccupied, where k_F is given by the condition:

$$n = \frac{k_F^3}{3\pi^2}. \quad (2.21)$$

This free and independent electron ground state is described by some rather unimaginative nomenclature:

The sphere of radius k_F (the *Fermi wave vector*) containing the occupied one electron levels is called the *Fermi sphere*.

The surface of the Fermi sphere, which separates the occupied from the unoccupied levels is called the *Fermi surface*. (We shall see, starting with Chapter 8, that the Fermi surface is one of the fundamental constructions in the modern theory of metals; in general it is not spherical.)

The momentum $\hbar k_F = p_F$ of the occupied one-electron levels of highest energy is known as the *Fermi momentum*; their energy, $\varepsilon_F = \hbar^2 k_F^2 / 2m$ is the *Fermi energy*; and their velocity, $v_F = p_F / m$, is the *Fermi velocity*. The Fermi velocity plays a role in the theory of metals comparable to the thermal velocity, $v = (3k_B T / m)^{1/2}$, in a classical gas.

All these quantities can be evaluated in terms of the conduction electron density, via Eq. (2.21). For estimating them numerically it is often more convenient to express them in terms of the dimensionless parameter r_s/a_0 (see page 4), which varies from about 2 to 6 in the metallic elements. Taken together, Eqs. (1.2) and (2.21) give

$$k_F = \frac{(9\pi/4)^{1/3}}{r_s} = \frac{1.92}{r_s}, \quad (2.22)$$

or

$$k_F = \frac{3.63}{r_s/a_0} \text{ \AA}^{-1}. \quad (2.23)$$

Since the Fermi wave vector is of the order of inverse angstroms, the de Broglie wavelength of the most energetic electrons is of the order of angstroms.

The Fermi velocity is

$$v_F = \left(\frac{\hbar}{m} \right) k_F = \frac{4.20}{r_s/a_0} \times 10^8 \text{ cm/sec}. \quad (2.24)$$

This is a substantial velocity (about 1 percent of the velocity of light). From the viewpoint of classical statistical mechanics this is quite a surprising result, for we are

describing the ground state ($T = 0$), and all particles in a classical gas have zero velocity at $T = 0$. Even at room temperature the thermal (i.e., average) velocity for a classical particle with the electronic mass is only of order 10^7 cm/sec.

The Fermi energy is conveniently written in the form (since $a_0 = \hbar^2 / me^2$)

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2m} = \left(\frac{e^2}{2a_0} \right) (k_F a_0)^2. \quad (2.25)$$

Here $e^2/2a_0$, known as the rydberg (Ry), is the ground-state binding energy of the hydrogen atom, 13.6 electron volts.¹³ The rydberg is as convenient a unit of atomic energies as the Bohr radius is of atomic distances. Since $k_F a_0$ is of the order of unity, Eq. (2.25) demonstrates that the Fermi energy has the magnitude of a typical atomic binding energy. Using (2.23) and $a_0 = 0.529 \times 10^{-8}$ cm, we find the explicit numerical form:

$$\varepsilon_F = \frac{50.1 \text{ eV}}{(r_s/a_0)^2}, \quad (2.26)$$

indicating a range of Fermi energies for the densities of metallic elements between 1.5 and 15 electron volts.

Table 2.1 lists the Fermi energy, velocity, and wave vector for the metals whose conduction electron densities are given in Table 1.1.

To calculate the ground-state energy of N electrons in a volume V we must add up the energies of all the one-electron levels inside the Fermi sphere¹⁴:

$$E = 2 \sum_{k < k_F} \frac{\hbar^2}{2m} k^2. \quad (2.27)$$

Quite generally, in summing any smooth function $F(\mathbf{k})$ over all allowed values of $\mathbf{k}$, one may proceed as follows:

Because the volume of k -space per allowed $\mathbf{k}$ value is $\Delta k = 8\pi^3/V$ (see Eq. (2.18)) it is convenient to write

$$\sum_{\mathbf{k}} F(\mathbf{k}) = \frac{V}{8\pi^3} \sum_{\mathbf{k}} F(\mathbf{k}) \Delta k, \quad (2.28)$$

for in the limit as $\Delta k \rightarrow 0$ (i.e., $V \rightarrow \infty$) the sum $\sum F(\mathbf{k}) \Delta k$ approaches the integral $\int d\mathbf{k} F(\mathbf{k})$, provided only that $F(\mathbf{k})$ does not vary appreciably¹⁵ over distances in k -space of order $2\pi/L$. We may therefore rearrange (2.28) and write

$$\lim_{V \rightarrow \infty} \frac{1}{V} \sum_{\mathbf{k}} F(\mathbf{k}) = \int \frac{d\mathbf{k}}{8\pi^3} F(\mathbf{k}). \quad (2.29)$$

In applying (2.29) to finite, but macroscopically large, systems one always assumes that $(1/V) \sum F(\mathbf{k})$ differs negligibly from its infinite volume limit (for example, one

¹³ Strictly speaking, the rydberg is the binding energy in the approximation of infinite proton mass. An electron volt is the energy gained by an electron crossing a potential of 1 volt; $1 \text{ eV} = 1.602 \times 10^{-12} \text{ erg} = 1.602 \times 10^{-19} \text{ joule}$.

¹⁴ The factor of 2 is for the two spin levels allowed for each $\mathbf{k}$.

¹⁵ The most celebrated case in which F fails to satisfy this condition is the condensation of the ideal Bose gas. In applications to metals the problem never arises.

Table 2.1
FERMI ENERGIES, FERMI TEMPERATURES, FERMI WAVE VECTORS, AND
FERMI VELOCITIES FOR REPRESENTATIVE METALS^a

ELEMENT	r_s/a_0	ε_F	T_F	k_F	v_F
Li	3.25	4.74 eV	5.51×10^4 K	1.12×10^8 cm ⁻¹	1.29×10^8 cm/sec
Na	3.93	3.24	3.77	0.92	1.07
K	4.86	2.12	2.46	0.75	0.86
Rb	5.20	1.85	2.15	0.70	0.81
Cs	5.62	1.59	1.84	0.65	0.75
Cu	2.67	7.00	8.16	1.36	1.57
Ag	3.02	5.49	6.38	1.20	1.39
Au	3.01	5.53	6.42	1.21	1.40
Be	1.87	14.3	16.6	1.94	2.25
Mg	2.66	7.08	8.23	1.36	1.58
Ca	3.27	4.69	5.44	1.11	1.28
Sr	3.57	3.93	4.57	1.02	1.18
Ba	3.71	3.64	4.23	0.98	1.13
Nb	3.07	5.32	6.18	1.18	1.37
Fe	2.12	11.1	13.0	1.71	1.98
Mn	2.14	10.9	12.7	1.70	1.96
Zn	2.30	9.47	11.0	1.58	1.83
Cd	2.59	7.47	8.68	1.40	1.62
Hg	2.65	7.13	8.29	1.37	1.58
Al	2.07	11.7	13.6	1.75	2.03
Ga	2.19	10.4	12.1	1.66	1.92
In	2.41	8.63	10.0	1.51	1.74
Tl	2.48	8.15	9.46	1.46	1.69
Sn	2.22	10.2	11.8	1.64	1.90
Pb	2.30	9.47	11.0	1.58	1.83
Bi	2.25	9.90	11.5	1.61	1.87
Sb	2.14	10.9	12.7	1.70	1.96

^a The table entries are calculated from the values of r_s/a_0 given in Table 1.1 using $m = 9.11 \times 10^{-28}$ grams.

assumes that the electronic energy per unit volume in a 1-cm cube of copper is the same as in a 2-cm cube).

Using (2.29) to evaluate (2.27), we find that the energy density of the electron gas is:

$$\frac{E}{V} = \frac{1}{4\pi^3} \int_{k < k_F} d\mathbf{k} \frac{\hbar^2 k^2}{2m} = \frac{1}{\pi^2} \frac{\hbar^2 k_F^5}{10m}. \quad (2.30)$$

To find the energy per electron, E/N , in the ground state, we must divide this by $N/V = k_F^3/3\pi^2$, which gives

$$\frac{E}{N} = \frac{3}{10} \frac{\hbar^2 k_F^2}{m} = \frac{3}{5} \varepsilon_F. \quad (2.31)$$

We can also write this result as

$$\frac{E}{N} = \frac{3}{5} k_B T_F \quad (2.32)$$

where T_F , the Fermi temperature, is

$$T_F = \frac{\varepsilon_F}{k_B} = \frac{58.2}{(r_s/a_0)^2} \times 10^4 \text{ K}. \quad (2.33)$$

Note, in contrast to this, that the energy per electron in a classical ideal gas, $\frac{3}{2} k_B T$, vanishes at $T = 0$ and achieves a value as large as (2.32) only at $T = \frac{2}{3} T_F \approx 10^4$ K.

Given the ground-state energy E , one can calculate the pressure exerted by the electron gas from the relation $P = -(\partial E / \partial V)_N$. Since $E = \frac{3}{5} N \varepsilon_F$ and ε_F is proportional to k_F^2 , which depends on V only through a factor $n^{2/3} = (N/V)^{2/3}$, it follows that¹⁶

$$P = \frac{2}{3} \frac{E}{V}. \quad (2.34)$$

One can also calculate the compressibility, K , or bulk modulus, $B = 1/K$, defined by:

$$B = \frac{1}{K} = -V \frac{\partial P}{\partial V}. \quad (2.35)$$

Since E is proportional to $V^{-2/3}$, Eq. (2.34) shows that P varies as $V^{-5/3}$, and therefore

$$B = \frac{5}{3} P = \frac{10}{9} \frac{E}{V} = \frac{2}{3} n \varepsilon_F \quad (2.36)$$

or

$$B = \left(\frac{6.13}{r_s/a_0} \right)^5 \times 10^{10} \text{ dynes/cm}^2. \quad (2.37)$$

In Table 2.2 we compare the free electron bulk moduli (2.37) calculated from r_s/a_0 , with the measured bulk moduli, for several metals. The agreement for the heavier alkali metals is fortuitously good, but even when (2.37) is substantially off, as it is in

Table 2.2
BULK MODULI IN 10^{10} DYNES/CM² FOR SOME
TYPICAL METALS^a

METAL	FREE ELECTRON B	MEASURED B
Li	23.9	11.5
Na	9.23	6.42
K	3.19	2.81
Rb	2.28	1.92
Cs	1.54	1.43
Cu	63.8	134.3
Ag	34.5	99.9
Al	228	76.0

^a The free electron value is that for a free electron gas at the observed density of the metal, as calculated from Eq. (2.37).

¹⁶ At nonzero temperatures the pressure and energy density continue to obey this relation. See (2.101).

the noble metals, it is still of about the right order of magnitude (though it varies from three times too large to three times too small, through the table). It is absurd to expect that the free electron gas pressure alone should completely determine the resistance of a metal to compression, but Table 2.2 demonstrates that this pressure is at least as important as any other effects.

THERMAL PROPERTIES OF THE FREE ELECTRON GAS: THE FERMI-DIRAC DISTRIBUTION

When the temperature is not zero it is necessary to examine the excited states of the N -electron system as well as its ground state, for according to the basic principles of statistical mechanics, if an N -particle system is in thermal equilibrium at temperature T , then its properties should be calculated by averaging over all N -particle stationary states, assigning to each state of energy E a weight $P_N(E)$ proportional to $e^{-E/k_B T}$:

$$P_N(E) = \frac{e^{-E/k_B T}}{\sum e^{-E_\alpha^N/k_B T}}. \quad (2.38)$$

(Here E_α^N is the energy of the α th stationary state of the N -electron system, the sum being over all such states.)

The denominator of (2.38) is known as the partition function, and is related to the Helmholtz free energy, $F = U - TS$ (where U is the internal energy and S , the entropy) by

$$\sum e^{-E_\alpha^N/k_B T} = e^{-F_N/k_B T}. \quad (2.39)$$

We can therefore write (2.38) more compactly as:

$$P_N(E) = e^{-(E - F_N)/k_B T}. \quad (2.40)$$

Because of the exclusion principle, to construct an N -electron state one must fill N different one-electron levels. Thus each N -electron stationary state can be specified by listing which of the N one-electron levels are filled in that state. A very useful quantity to know is f_i^N , the probability of there being an electron in the particular one-electron level i , when the N -electron system is in thermal equilibrium.¹⁷ This probability is simply the sum of the independent probabilities of finding the N -electron system in any one of those N -electron states in which the i th level is occupied:

$$f_i^N = \sum P_N(E_\alpha^N) \quad \begin{array}{l} \text{(summation over all } N\text{-electron} \\ \text{states } \alpha \text{ in which there is an elec-} \\ \text{tron in the one-electron level } i). \end{array} \quad (2.41)$$

We can evaluate f_i^N by the following three observations:

1. Since the probability of an electron being in the level i is just one minus the probability of no electron being in the level i (those being the only two possibilities

allowed by the exclusion principle) we could equally well write (2.41) as

$$f_i^N = 1 - \sum P_N(E_\gamma^N) \quad \begin{array}{l} \text{(summation over all } N\text{-electron} \\ \text{states } \gamma \text{ in which there is no elec-} \\ \text{tron in the one-electron level } i). \end{array} \quad (2.42)$$

2. By taking any $(N + 1)$ -electron state in which there is an electron in the one-electron level i , we can construct an N -electron state in which there is no electron in the level i , by simply removing the electron in the i th level, leaving the occupation of all the other levels unaltered. Furthermore, any N -electron state with no electron in the one-electron level i can be so constructed from just one $(N + 1)$ -electron state with an electron in the level i .¹⁸ Evidently the energies of any N -electron state and the corresponding $(N + 1)$ -electron state differ by just ϵ_i , the energy of the only one-electron level whose occupation is different in the two states. Thus the set of energies of all N -electron states with the level i unoccupied is the same as the set of energies of all $(N + 1)$ -electron states with the level i occupied, provided that each energy in the latter set is reduced by ϵ_i . We can therefore rewrite (2.42) in the peculiar form

$$f_i^N = 1 - \sum P_N(E_\alpha^{N+1} - \epsilon_i) \quad \begin{array}{l} \text{(summation over all } (N + 1)\text{-electron} \\ \text{states } \alpha \text{ in which there is an electron} \\ \text{in the one-electron level } i). \end{array} \quad (2.43)$$

But Eq. (2.40) permits us to write the summand as

$$P_N(E_\alpha^{N+1} - \epsilon_i) = e^{(\epsilon_i - \mu)/k_B T} P_{N+1}(E_\alpha^{N+1}), \quad (2.44)$$

where μ , known as the chemical potential, is given at temperature T by

$$\mu = F_{N+1} - F_N. \quad (2.45)$$

Substituting this into (2.43), we find:

$$f_i^N = 1 - e^{(\epsilon_i - \mu)/k_B T} \sum P_{N+1}(E_\alpha^{N+1}) \quad \begin{array}{l} \text{(summation over all } (N + 1)\text{-electron} \\ \text{states } \alpha \text{ in which there is an electron} \\ \text{in the one-electron level } i). \end{array} \quad (2.46)$$

Comparing the summation in (2.46) with that in (2.41) one finds that (2.46) simply asserts that

$$f_i^N = 1 - e^{(\epsilon_i - \mu)/k_B T} f_i^{N+1}. \quad (2.47)$$

3. Equation (2.47) gives an exact relation between the probability of the one-electron level i being occupied at temperature T in an N -electron system, and in an $(N + 1)$ -electron system. When N is very large (and we are typically interested in N of the order of 10^{22}) it is absurd to imagine that by the addition of a single extra electron we could appreciably alter this probability for more than an insignificant handful of one-electron levels.¹⁹ We may therefore replace f_i^{N+1} by f_i^N in (2.47), which

¹⁸ Namely the one obtained by occupying all those levels occupied in the N -electron state plus the i th level.

¹⁹ For a typical level, changing N by one alters the probability of occupation by order $1/N$. See Problem 4.

¹⁷ In the case we are interested in the level i is specified by the electron's wave vector $\mathbf{k}$ and the projection s of its spin along some axis.

makes it possible to solve for f_i^N :

$$f_i^N = \frac{1}{e^{(\varepsilon_i - \mu)/k_B T} + 1}. \quad (2.48)$$

In subsequent formulas we shall drop the explicit reference to the N dependence of f_i , which is, in any event, carried through the chemical potential μ ; see (2.45). The value of N can always be computed, given the f_i , by noting that f_i is the mean number of electrons in the one-electron level²⁰ i . Since the total number of electrons N is just the sum over all levels of the mean number in each level,

$$N = \sum_i f_i = \sum_i \frac{1}{e^{(\varepsilon_i - \mu)/k_B T} + 1}, \quad (2.49)$$

which determines N as a function of the temperature T and chemical potential μ . In many applications, however, it is the temperature and N (or rather the density, $n = N/V$) that are given. In such cases (2.49) is used to determine the chemical potential μ as a function of n and T , permitting it to be eliminated from subsequent formulas in favor of the temperature and density. However the chemical potential is of considerable thermodynamic interest in its own right. Some of its important properties are summarized in Appendix B.²¹

THERMAL PROPERTIES OF THE FREE ELECTRON GAS: APPLICATIONS OF THE FERMI-DIRAC DISTRIBUTION

In a gas of free and independent electrons the one-electron levels are specified by the wave vector $\mathbf{k}$ and spin quantum number s , with energies that are independent of s (in the absence of a magnetic field) and given by Eq. (2.7); i.e.,

$$\varepsilon(\mathbf{k}) = \frac{\hbar^2 k^2}{2m}. \quad (2.50)$$

We first verify that the distribution function (2.49) is consistent with the ground-state ($T = 0$) properties derived above. In the ground state those and only those levels are occupied with $\varepsilon(\mathbf{k}) \leq \varepsilon_F$, so the ground-state distribution function must be

$$\begin{aligned} f_{\mathbf{k}s} &= 1, & \varepsilon(\mathbf{k}) < \varepsilon_F; \\ &= 0, & \varepsilon(\mathbf{k}) > \varepsilon_F. \end{aligned} \quad (2.51)$$

²⁰ *Proof:* A level can contain either 0 or 1 electron (more than one being prohibited by the exclusion principle). The mean number of electrons is therefore 1 times the probability of 1 electron plus 0 times the probability of 0 electrons. Thus the mean number of electrons in the level is numerically equal to the probability of its being occupied. Note that this would not be so if multiple occupation of levels were permitted.

²¹ The chemical potential plays a more fundamental role when the distribution (2.48) is derived in the grand canonical ensemble. See, for example, F. Reif, *Statistical and Thermal Physics*, McGraw-Hill, New York, 1965, p. 350. Our somewhat unorthodox derivation, which can also be found in Reif, uses only the canonical ensemble.

On the other hand, as $T \rightarrow 0$, the limiting form of the Fermi-Dirac distribution (2.48) is

$$\lim_{T \rightarrow 0} f_{\mathbf{k}s} = \begin{cases} 1, & \varepsilon(\mathbf{k}) < \mu; \\ 0, & \varepsilon(\mathbf{k}) > \mu. \end{cases} \quad (2.52)$$

For these to be consistent it is necessary that

$$\lim_{T \rightarrow 0} \mu = \varepsilon_F. \quad (2.53)$$

We shall see shortly that for metals the chemical potential remains equal to the Fermi energy to a high degree of precision, all the way up to room temperature. As a result, people frequently fail to make any distinction between the two when dealing with metals. This, however, can be dangerously misleading. In precise calculations it is essential to keep track of the extent to which μ , the chemical potential, differs from its zero temperature value, ε_F .

The most important single application of Fermi-Dirac statistics is the calculation of the electronic contribution to the constant-volume specific heat of a metal,

$$c_v = \frac{T}{V} \left(\frac{\partial S}{\partial T} \right)_V = \left(\frac{\partial u}{\partial T} \right)_V, \quad u = \frac{U}{V}. \quad (2.54)$$

In the independent electron approximation the internal energy U is just the sum over one-electron levels of $\varepsilon(\mathbf{k})$ times the mean number of electrons in the level²²:

$$U = 2 \sum_{\mathbf{k}} \varepsilon(\mathbf{k}) f(\varepsilon(\mathbf{k})). \quad (2.55)$$

We have introduced the *Fermi function* $f(\varepsilon)$ to emphasize that $f_{\mathbf{k}}$ depends on $\mathbf{k}$ only through the electronic energy $\varepsilon(\mathbf{k})$:

$$f(\varepsilon) = \frac{1}{e^{(\varepsilon - \mu)/k_B T} + 1}. \quad (2.56)$$

If we divide both sides of (2.55) by the volume V , then (2.29) permits us to write the energy density $u = U/V$ as

$$u = \int \frac{d\mathbf{k}}{4\pi^3} \varepsilon(\mathbf{k}) f(\varepsilon(\mathbf{k})). \quad (2.57)$$

If we also divide both sides of (2.49) by V , then we can supplement (2.57) by an equation for the electronic density $n = N/V$, and use it to eliminate the chemical potential:

$$n = \int \frac{d\mathbf{k}}{4\pi^3} f(\varepsilon(\mathbf{k})). \quad (2.58)$$

In evaluating integrals like (2.57) and (2.58) of the form

$$\int \frac{d\mathbf{k}}{4\pi^3} F(\varepsilon(\mathbf{k})), \quad (2.59)$$

²² As usual, the factor of 2 reflects the fact that each k -level can contain two electrons of opposite spin orientations.

one often exploits the fact that the integrand depends on $\mathbf{k}$ only through the electronic energy $\varepsilon = \hbar^2 k^2 / 2m$, by evaluating the integral in spherical coordinates and changing variables from k to ε :

$$\int \frac{d\mathbf{k}}{4\pi^3} F(\varepsilon(\mathbf{k})) = \int_0^\infty \frac{k^2 dk}{\pi^2} F(\varepsilon(\mathbf{k})) = \int_{-\infty}^\infty d\varepsilon g(\varepsilon) F(\varepsilon). \quad (2.60)$$

Here

$$g(\varepsilon) = \frac{m}{\hbar^2 \pi^2} \sqrt{\frac{2m\varepsilon}{\hbar^2}}, \quad \varepsilon > 0; \\ = 0, \quad \varepsilon < 0. \quad (2.61)$$

Since the integral (2.59) is an evaluation of $(1/V) \sum_{\mathbf{k}s} F(\varepsilon(\mathbf{k}))$, the form in (2.60) shows that

$$g(\varepsilon) d\varepsilon = \left(\frac{1}{V} \right) \times \left[\begin{array}{l} \text{the number of one-electron} \\ \text{levels in the energy range} \\ \text{from } \varepsilon \text{ to } \varepsilon + d\varepsilon \end{array} \right]. \quad (2.62)$$

For this reason $g(\varepsilon)$ is known as the density of levels per unit volume (or often simply as the density of levels). A dimensionally more transparent way of writing g is

$$g(\varepsilon) = \frac{3}{2} \frac{n}{\varepsilon_F} \left(\frac{\varepsilon}{\varepsilon_F} \right)^{1/2}, \quad \varepsilon > 0; \\ = 0, \quad \varepsilon < 0, \quad (2.63)$$

where ε_F and k_F are defined by the zero-temperature equations (2.21) and (2.25). A quantity of particular numerical importance is the density of levels at the Fermi energy, which (2.61) and (2.63) give in either of the two equivalent forms:

$$g(\varepsilon_F) = \frac{mk_F}{\hbar^2 \pi^2} \quad (2.64)$$

or

$$g(\varepsilon_F) = \frac{3}{2} \frac{n}{\varepsilon_F}. \quad (2.65)$$

Using this notation, we rewrite (2.57) and (2.58) as:

$$u = \int_{-\infty}^\infty d\varepsilon g(\varepsilon) \varepsilon f(\varepsilon) \quad (2.66)$$

and

$$n = \int_{-\infty}^\infty d\varepsilon g(\varepsilon) f(\varepsilon). \quad (2.67)$$

We do this both for notational simplicity and because in this form the free electron approximation enters only through the particular evaluation (2.61) or (2.63) of the density of levels g . We can define a density of levels, via (2.62), in terms of which (2.66)

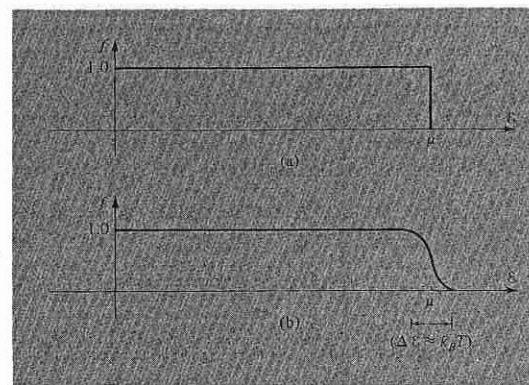


Figure 2.3

The Fermi function, $f(\varepsilon) = 1/[e^{\beta(\varepsilon - \mu)} + 1]$ versus ε for given μ , at (a) $T = 0$ and (b) $T \approx 0.01\mu$ (of order room temperature, at typical metallic densities). The two curves differ only in a region of order $k_B T$ about μ .

and (2.67) remain valid for any set of noninteracting (that is, independent) electrons.²³ Thus we shall later be able to apply results deduced from (2.66) and (2.67) to considerably more sophisticated models of independent electrons in metals.

In general, the integrals (2.66) and (2.67) have a rather complex structure. There is, however, a simple systematic expansion that exploits the fact that at almost all temperatures of interest in metals, T is very much smaller than the Fermi temperature (2.33). In Figure 2.3 the Fermi function $f(\varepsilon)$ is plotted at $T = 0$ and at room temperature for typical metallic densities ($k_B T/\mu \approx 0.01$). Evidently f differs from its zero temperature form only in a small region about μ of width a few $k_B T$. Thus the way in which integrals of the form $\int_{-\infty}^\infty H(\varepsilon) f(\varepsilon) d\varepsilon$ differ from their zero temperature values, $\int_{-\infty}^\infty H(\varepsilon) d\varepsilon$, will be entirely determined by the form of $H(\varepsilon)$ near $\varepsilon = \mu$. If $H(\varepsilon)$ does not vary rapidly in the energy range of the order of $k_B T$ about μ , the temperature dependence of the integral should be given quite accurately by replacing $H(\varepsilon)$ by the first few terms in its Taylor expansion about $\varepsilon = \mu$:

$$H(\varepsilon) = \sum_{n=0}^{\infty} \frac{d^n}{d\varepsilon^n} H(\varepsilon) \Big|_{\varepsilon=\mu} \frac{(\varepsilon - \mu)^n}{n!}. \quad (2.68)$$

This procedure is carried out in Appendix C. The result is a series of the form:

$$\int_{-\infty}^\infty H(\varepsilon) f(\varepsilon) d\varepsilon = \int_{-\infty}^\mu H(\varepsilon) d\varepsilon + \sum_{n=1}^{\infty} (k_B T)^{2n} a_n \frac{d^{2n-1}}{d\varepsilon^{2n-1}} H(\varepsilon) \Big|_{\varepsilon=\mu} \quad (2.69)$$

which is known as the Sommerfeld expansion.²⁴ The a_n are dimensionless constants of the order of unity. The functions H one typically encounters have major variations on an energy scale of the order of μ , and generally $(d/d\varepsilon)^n H(\varepsilon)|_{\varepsilon=\mu}$ is of the order of $H(\mu)/\mu^n$. When this is the case, successive terms in the Sommerfeld expansion are

²³ See Chapter 8.

²⁴ The expansion is not always exact, but is highly reliable unless $H(\varepsilon)$ has a singularity very close to $\varepsilon = \mu$. If, for example, H is singular at $\varepsilon = 0$ (as is the free electron density of levels (2.63)) then the expansion will neglect terms of the order of $\exp(-\mu/k_B T)$, which are typically of order $e^{-100} \sim 10^{-43}$. See also Problem 1.

smaller by $O(k_B T/\mu)^2$ which is $O(10^{-4})$ at room temperature. Consequently, in actual calculations only the first and (very occasionally) the second terms are retained in the sum in (2.69). The explicit form for these is (Appendix C):

$$\int_{-\infty}^{\infty} H(\varepsilon) f(\varepsilon) d\varepsilon = \int_{-\infty}^{\mu} H(\varepsilon) d\varepsilon + \frac{\pi^2}{6} (k_B T)^2 H'(\mu) + \frac{7\pi^4}{360} (k_B T)^4 H'''(\mu) + O\left(\frac{k_B T}{\mu}\right)^6. \quad (2.70)$$

To evaluate the specific heat of a metal at temperatures small compared with T_F we apply the Sommerfeld expansion (2.70) to the electronic energy and number densities (Eqs. (2.66) and (2.67)):

$$u = \int_{-\infty}^{\mu} \varepsilon g(\varepsilon) d\varepsilon + \frac{\pi^2}{6} (k_B T)^2 [\mu g'(\mu) + g(\mu)] + O(T^4), \quad (2.71)$$

$$n = \int_{-\infty}^{\mu} g(\varepsilon) d\varepsilon + \frac{\pi^2}{6} (k_B T)^2 g'(\mu) + O(T^4). \quad (2.72)$$

Equation (2.72) as we shall presently see in detail, implies that μ differs from its $T = 0$ value, ε_F , by terms of order T^2 . Thus, correctly to order T^2 , we may write

$$\int_0^{\mu} H(\varepsilon) d\varepsilon = \int_0^{\varepsilon_F} H(\varepsilon) d\varepsilon + (\mu - \varepsilon_F) H(\varepsilon_F). \quad (2.73)$$

If we apply this expansion to the integrals in (2.71) and (2.72), and replace μ by ε_F in the terms already of order T^2 in these equations, we find

$$u = \int_0^{\varepsilon_F} \varepsilon g(\varepsilon) d\varepsilon + \varepsilon_F \left\{ (\mu - \varepsilon_F) g(\varepsilon_F) + \frac{\pi^2}{6} (k_B T)^2 g'(\varepsilon_F) \right\} + \frac{\pi^2}{6} (k_B T)^2 g(\varepsilon_F) + O(T^4), \quad (2.74)$$

$$n = \int_0^{\varepsilon_F} g(\varepsilon) d\varepsilon + \left\{ (\mu - \varepsilon_F) g(\varepsilon_F) + \frac{\pi^2}{6} (k_B T)^2 g'(\varepsilon_F) \right\}. \quad (2.75)$$

The temperature-independent first terms on the right sides of (2.74) and (2.75) are just the values of u and n in the ground state. Since we are calculating the specific heat at constant density, n is independent of temperature, and (2.75) reduces to

$$0 = (\mu - \varepsilon_F) g(\varepsilon_F) + \frac{\pi^2}{6} (k_B T)^2 g'(\varepsilon_F), \quad (2.76)$$

which determines the deviation of the chemical potential from ε_F :

$$\mu = \varepsilon_F - \frac{\pi^2}{6} (k_B T)^2 \frac{g'(\varepsilon_F)}{g(\varepsilon_F)}. \quad (2.77)$$

Since for free electrons $g(\varepsilon)$ varies as $\varepsilon^{1/2}$ (see Eq. (2.63)), this gives

$$\mu = \varepsilon_F \left[1 - \frac{1}{3} \left(\frac{\pi k_B T}{2\varepsilon_F} \right)^2 \right], \quad (2.78)$$

which is, as we asserted above, a shift of the order of T^2 and typically only about 0.01 percent, even at room temperature.

Equation (2.76) sets the term in braces in (2.74) equal to zero, thereby simplifying the form of the thermal energy density at constant electronic density:

$$u = u_0 + \frac{\pi^2}{6} (k_B T)^2 g(\varepsilon_F) \quad (2.79)$$

where u_0 is the energy density in the ground state. The specific heat of the electron gas is therefore

$$c_v = \left(\frac{\partial u}{\partial T} \right)_n = \frac{\pi^2}{3} k_B^2 T g(\varepsilon_F) \quad (2.80)$$

or, for free electrons (see (2.65)),

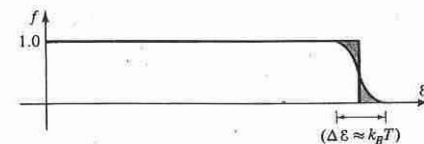
$$c_v = \frac{\pi^2}{2} \left(\frac{k_B T}{\varepsilon_F} \right) n k_B. \quad (2.81)$$

Comparing this with the classical result for an ideal gas, $c_v = 3nk_B/2$, we see that the effect of Fermi-Dirac statistics is to depress the specific heat by a factor of $(\pi^2/3)(k_B T/\varepsilon_F)$, which is proportional to the temperature, and even at room temperature is only of order 10^{-2} . This explains the absence of any observable contribution of the electronic degrees of freedom to the specific heat of a metal at room temperature.

If one is willing to dispense with the precise numerical coefficient, one can understand this behavior of the specific heat quite simply from the temperature dependence of the Fermi function itself. The increase in energy of the electrons when the temperature is raised from $T = 0$ comes about entirely because some electrons with energies within $O(k_B T)$ below ε_F (the darkly shaded region of Figure 2.4) have been excited to an energy range of $O(k_B T)$ above ε_F (the lightly shaded region of Figure 2.4). The number of electrons per unit volume that have been so excited is the width, $k_B T$, of the energy interval times the density of levels per unit volume $g(\varepsilon_F)$. Furthermore, the excitation energy is of order $k_B T$, and hence the total thermal energy density is of order $g(\varepsilon_F)(k_B T)^2$ above the ground-state energy. This misses the precise result (2.79) by a factor of $\pi^2/6$, but it gives a simple physical picture, and is useful for rough estimates.

Figure 2.4

The Fermi function at nonzero T . The distribution differs from its $T = 0$ form because some electrons just below ε_F (darkly shaded region) have been excited to levels just above ε_F (lightly shaded region).



The prediction of a linear specific heat is one of the most important consequences of Fermi-Dirac statistics, and provides a further simple test of the electron gas theory of a metal, provided one can be sure that degrees of freedom other than the electronic ones do not make comparable or even bigger contributions. As it happens, the ionic degrees of freedom completely dominate the specific heat at high temperatures. However, well below room temperature their contribution falls off as the cube of the

Table 2.3

SOME ROUGH EXPERIMENTAL VALUES FOR THE COEFFICIENT OF THE LINEAR TERM IN T OF THE MOLAR SPECIFIC HEATS OF METALS, AND THE VALUES GIVEN BY SIMPLE FREE ELECTRON THEORY

ELEMENT	FREE ELECTRON γ (in 10^{-4} cal-mole $^{-1}$ -K $^{-2}$)	MEASURED γ	RATIO ^a (m^*/m)
Li	1.8	4.2	2.3
Na	2.6	3.5	1.3
K	4.0	4.7	1.2
Rb	4.6	5.8	1.3
Cs	5.3	7.7	1.5
Cu	1.2	1.6	1.3
Ag	1.5	1.6	1.1
Au	1.5	1.6	1.1
Be	1.2	0.5	0.42
Mg	2.4	3.2	1.3
Ca	3.6	6.5	1.8
Sr	4.3	8.7	2.0
Ba	4.7	6.5	1.4
Nb	1.6	20	12
Fe	1.5	12	8.0
Mn	1.5	40	27
Zn	1.8	1.4	0.78
Cd	2.3	1.7	0.74
Hg	2.4	5.0	2.1
Al	2.2	3.0	1.4
Ga	2.4	1.5	0.62
In	2.9	4.3	1.5
Tl	3.1	3.5	1.1
Sn	3.3	4.4	1.3
Pb	3.6	7.0	1.9
Bi	4.3	0.2	0.047
Sb	3.9	1.5	0.38

^a Since the theoretical value of γ is proportional to the density of levels at the Fermi level, which in turn is proportional to the electronic mass m , one sometimes defines a specific heat effective mass m^* so that m^*/m is the ratio of the measured γ to the free electron γ . Beware of identifying this specific heat effective mass with any of the many other effective masses used in solid-state theory. (See, for example, the index entries under "effective mass.")

temperature (Chapter 23), and at very low temperatures it drops below the electronic contribution, which only decreases linearly with T . In order to separate out these two contributions it has become the practice to plot c_v/T against T^2 , for if the electronic and ionic contributions together result in the low-temperature form,

$$c_v = \gamma T + AT^3, \quad (2.82)$$

then

$$\frac{c_v}{T} = \gamma + AT^2. \quad (2.83)$$

One can thus find γ by extrapolating the c_v/T curve linearly down to $T^2 = 0$, and noting where it intercepts the c_v/T -axis. Measured metallic specific heats typically contain a linear term that becomes comparable to the cubic one at a few degrees Kelvin.²⁵

Specific heat data are usually quoted in joules (or calories) per mole per degree Kelvin. Since a mole of free electron metal contains ZN_A conduction electrons (where Z is the valence and N_A is Avogadro's number) and occupies a volume ZN_A/n , we must multiply the heat capacity per unit volume, c_v by ZN_A/n , in order to get the heat capacity per mole, C :

$$C = \frac{\pi^2}{3} ZR \frac{k_B T g(\epsilon_F)}{n}, \quad (2.84)$$

where $R = k_B N_A = 8.314$ joules/mole = 1.99 calories/mole-K. Using the free electron density of levels (2.65) and the evaluation (2.33) of ϵ_F/k_B , we find a free electron contribution to the heat capacity per mole of $C = \gamma T$, where

$$\gamma = \frac{1}{2} \pi^2 R \frac{Z}{T_F} = 0.169 Z \left(\frac{r_s}{a_0} \right)^2 \times 10^{-4} \text{ cal-mole}^{-1} \text{ K}^{-2}. \quad (2.85)$$

Some rough, measured values of γ are displayed in Table 2.3, together with the free electron values implied by (2.85) and the values of r_s/a_0 in Table 1.1. Note that the alkali metals continue to be reasonably well described by free electron theory, as do the noble metals (Cu, Ag, Au). Note also, however, the striking disparities in Fe and Mn (experiment of the order of 10 times theory) as well as those in Bi and Sb (experiment of the order of 0.1 times theory). These large deviations are now qualitatively understood on fairly general grounds, and we shall return to them in Chapter 15.

THE SOMMERFELD THEORY OF CONDUCTION IN METALS

To find the velocity distribution for electrons in metals, consider a small²⁶ volume element of k -space about a point $\mathbf{k}$, of volume $d\mathbf{k}$. Allowing for the twofold spin

²⁵ Since constant density is hard to arrange experimentally, one generally measures the specific heat at constant pressure, c_p . However, one can show (Problem 2) that for the metallic free electron gas at room temperature and below, $c_p/c_v = 1 + O(k_B T/\epsilon_F)^2$. Thus at temperatures where the electronic contribution to the specific heat becomes observable (a few degrees Kelvin) the two specific heats differ by a negligible amount.

²⁶ Small enough that the Fermi function and other functions of physical interest vary negligibly throughout the volume element, but large enough that it contains very many one-electron levels.

degeneracy, the number of one-electron levels in this volume element is (see (2.18))

$$\left(\frac{V}{4\pi^3}\right) dk. \quad (2.86)$$

The probability of each level being occupied is just $f(\varepsilon(\mathbf{k}))$, and therefore the total number of electrons in the k -space volume element is

$$\frac{V}{4\pi^3} f(\varepsilon(\mathbf{k})) dk, \quad \varepsilon(\mathbf{k}) = \frac{\hbar^2 \mathbf{k}^2}{2m}. \quad (2.87)$$

Since the velocity of a free electron with wave vector $\mathbf{k}$ is $\mathbf{v} = \hbar \mathbf{k}/m$ (Eq. (2.12)), the number of electrons in an element of volume $d\mathbf{v}$ about $\mathbf{v}$ is the same as the number in an element of volume $d\mathbf{k} = (m/\hbar)^3 d\mathbf{v}$ about $\mathbf{k} = m\mathbf{v}/\hbar$. Consequently the total number of electrons per unit volume of real space in a velocity space element of volume $d\mathbf{v}$ about $\mathbf{v}$ is

$$f(\mathbf{v}) d\mathbf{v}, \quad (2.88)$$

where

$$f(\mathbf{v}) = \frac{(m/\hbar)^3}{4\pi^3} \frac{1}{\exp[(\frac{1}{2}mv^2 - \mu)/k_B T] + 1}. \quad (2.89)$$

Sommerfeld reexamined the Drude model, replacing the classical Maxwell-Boltzmann velocity distribution (2.1) by the Fermi-Dirac distribution (2.89). Using a velocity distribution constructed from quantum-mechanical arguments in an otherwise classical theory requires some justification.²⁷ One can describe the motion of an electron classically if one can specify its position and momentum as accurately as necessary, without violating the uncertainty principle.²⁸

A typical electron in a metal has a momentum of the order of $\hbar k_F$, so the uncertainty in its momentum, Δp , must be small compared with $\hbar k_F$ for a good classical description. Since, from (2.22), $k_F \sim 1/r_s$, the uncertainty in position must satisfy

$$\Delta x \sim \frac{\hbar}{\Delta p} \gg \frac{1}{k_F} \sim r_s, \quad (2.90)$$

where, from (1.2), r_s is of the order of the mean interelectronic distance—i.e., angstroms. Thus a classical description is impossible if one has to consider electrons localized to within atomic distances (also of the order of angstroms). However, the conduction electrons in a metal are not bound to particular ions, but can wander freely through the volume of the metal. In a macroscopic specimen, for most purposes there is no need to specify their position to an accuracy of 10^{-8} cm. The Drude model assumes a knowledge of the position of an electron primarily in only the two following contexts:

²⁷ A detailed analytical justification is fairly complicated to construct, just as it is a fairly subtle matter to specify with generality and precision when the quantum theory can be replaced by its classical limit. The underlying physics, however, is straightforward.

²⁸ There is also a somewhat more specialized limitation on the use of classical mechanics in describing conduction electrons. The energy of motion of an electron in the plane perpendicular to a uniform applied magnetic field is quantized in multiples of $\hbar\omega_c$ (Chapter 14). Even for fields as large as 10^4 gauss, this is a very small energy, but in suitably prepared samples at temperatures of a few degrees Kelvin, these quantum effects become observable, and are, in fact, of great practical importance.

1. When spatially varying electromagnetic fields or temperature gradients are applied, one must be able to specify the position of an electron on a scale small compared with the distance λ over which the fields or temperature gradients vary. For most applications the applied fields or temperature gradients do not vary appreciably on the scale of angstroms, and the necessary precision of definition in the electron's position need not lead to an unacceptably large uncertainty in its momentum. For example, the electric field associated with visible light varies appreciably only over a distance of order 10^3 Å. If, however, the wave length is very much shorter than this (for example, X rays), one must use quantum mechanics to describe the electronic motion induced by the field.
2. There is also an implicit assumption in the Drude model that one can localize an electron to within substantially less than a mean free path ℓ , and one should therefore be suspicious of classical arguments when mean free paths much shorter than tens of angstroms occur. Fortunately, as we shall see below, mean free paths in metals are of the order of 100 Å at room temperature, and become longer still as the temperature drops.

There is thus a wide range of phenomena in which the behavior of a metallic electron is well described by classical mechanics. It is not, however, immediately evident from this that the behavior of N such electrons can be described by classical mechanics. Since the Pauli exclusion principle so profoundly affects the statistics of N electrons, why should it not have similarly drastic effects on their dynamics? That it does not follows from an elementary theorem, which we state without proof, since the proof, though simple, is notationally rather cumbersome:

Consider a system of N electrons whose interactions with one another are ignored, and which are exposed to an arbitrary space- and time-dependent electromagnetic field. Let the N -electron state at time 0 be formed by occupying a particular group of N one-electron levels, $\psi_1(0), \dots, \psi_N(0)$. Let $\psi_j(t)$ be the level $\psi_j(0)$ would evolve into in time t under the influence of the electromagnetic field if there were only a single electron present, which was in the level $\psi_j(0)$ at time zero. Then the correct N -electron state at time t will be the one formed by occupying the set of N one-electron levels $\psi_1(t), \dots, \psi_N(t)$.

Thus the dynamical behavior of N noninteracting electrons is completely determined by considering N independent one-electron problems. In particular, if the classical approximation is valid for each of these one-electron problems, it will also be valid for the whole N -electron system.²⁹

The use of Fermi-Dirac statistics affects only those predictions of the Drude model that require some knowledge of the electronic velocity distribution for their evaluation. If the rate $1/\tau$ at which an electron experiences collisions does not depend on its energy, then only our estimate of the electronic mean free path and our calculation of the thermal conductivity and thermopower are at all affected by a change in the equilibrium distribution function.

²⁹ Note that this implies that any classical configuration consistent with the exclusion principle at time $t = 0$ (i.e., having less than one electron of each spin per unit volume, in any momentum space region of volume $d\mathbf{p} = (2\pi\hbar)^3/V$) will remain consistent with the exclusion principle at all future times. This result can also be proved by purely classical reasoning as a direct corollary of Liouville's theorem. See Chapter 12.

Mean Free Path Using v_F (Eq. (2.24)) as a measure of the typical electronic speed, we can evaluate the mean free path $\ell = v_F \tau$ from Eq. (1.8) as follows:

$$\ell = \frac{(r_s/a_0)^2}{\rho_\mu} \times 92 \text{ \AA}. \quad (2.91)$$

Since the resistivity in microhm centimeters, ρ_μ , is typically 1 to 100 at room temperature, and since r_s/a_0 is typically 2 to 6, mean free paths of order a hundred angstroms long are possible even at room temperature.³⁰

Thermal Conductivity We continue to estimate the thermal conductivity by Eq. (1.51):

$$\kappa = \frac{1}{3} v^2 \tau c_v. \quad (2.92)$$

The correct specific heat (2.81) is smaller than the classical guess of Drude by a factor of order $k_B T/\epsilon_F$; the correct estimate of v^2 is not the classical thermal mean square velocity of order $k_B T/m$, but $v_F^2 = 2\epsilon_F/m$, which is larger than the classical value by a factor of order $\epsilon_F/k_B T$. Inserting these values in (2.92) and eliminating the relaxation time in favor of the conductivity through (1.6), we find

$$\frac{\kappa}{\sigma T} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = 2.44 \times 10^{-8} \text{ watt-ohm/K}^2. \quad (2.93)$$

This is remarkably close to Drude's fortuitously good value, thanks to the two compensating corrections of order $k_B T/\epsilon_F$, and in excellent agreement with the data in Table 1.6. We shall see (Chapter 13) that this value of the Lorenz number is a far better one than the very rough derivation of (2.93) would suggest.

Thermopower Drude's overestimate of the thermopower is also resolved by the use of Fermi-Dirac statistics. Substituting the specific heat, from Eq. (2.81), into Eq. (1.59) we find

$$Q = -\frac{\pi^2}{6} \frac{k_B}{e} \left(\frac{k_B T}{\epsilon_F} \right) = -1.42 \left(\frac{k_B T}{\epsilon_F} \right) \times 10^{-4} \text{ volt/K}, \quad (2.94)$$

which is smaller than Drude's estimate (Eq. (1.60)) by $O(k_B T/\epsilon_F) \sim 0.01$ at room temperature.

Other Properties Since the form of the electronic velocity distribution did not play a role in the calculation of the DC or AC conductivities, the Hall coefficient, or the magnetoresistance, the estimates given in Chapter 1 remain the same whether one uses Maxwell-Boltzmann or Fermi-Dirac statistics.

This is not the case, however, if one uses an energy-dependent relaxation time. If, for example, one thought the electrons collided with fixed scattering centers, then it would be natural to take an energy-independent mean free path, and hence a relaxation time $\tau = \ell/v \sim \ell/\epsilon^{1/2}$. Shortly after Drude set forth the electron gas model of a

³⁰ It is perhaps just as well that Drude estimated ℓ using the very much lower classical thermal velocity, or he might have been sufficiently baffled by such long mean free paths to abandon further investigation.

metal, H. A. Lorentz showed, using the classical Maxwell-Boltzmann velocity distribution, that an energy-dependent relaxation time would lead to temperature dependence in the DC and AC conductivities, as well as to a nonvanishing magnetoresistance and a field- and temperature-dependent Hall coefficient. As one might now expect from the inappropriateness of the classical velocity distribution, none of these corrections were in any way able to bring the discrepancies of the Drude model into better alignment with the observed facts about metals.³¹ Furthermore, we shall see (Chapter 13) that when the correct Fermi-Dirac velocity distribution is used, adding an energy dependence to the relaxation time has little significant effect on most of the quantities of interest in a metal.³² If one calculates the DC or AC conductivities, the magnetoresistance, or the Hall coefficient assuming an energy-dependent $\tau(\epsilon)$, the results one finds are the same as those one would have calculated assuming an energy-independent τ , equal to $\tau(\epsilon_F)$. In metals these quantities are determined almost entirely by the way in which electrons near the Fermi level are scattered.³³ This is another very important consequence of the Pauli exclusion principle, the justification of which will be given in Chapter 13.

PROBLEMS

1. The Free and Independent Electron Gas in Two Dimensions

- What is the relation between n and k_F in two dimensions?
- What is the relation between k_F and r_s in two dimensions?
- Prove that in two dimensions the free electron density of levels $g(\epsilon)$ is a constant independent of ϵ for $\epsilon > 0$, and 0 for $\epsilon < 0$. What is the constant?
- Show that because $g(\epsilon)$ is constant, every term in the Sommerfeld expansion for n vanishes except the $T = 0$ term. Deduce that $\mu = \epsilon_F$ at any temperature.
- Deduce from (2.67) that when $g(\epsilon)$ is as in (c), then

$$\mu + k_B T \ln(1 + e^{-\mu/k_B T}) = \epsilon_F. \quad (2.95)$$

- Estimate from (2.95) the amount by which μ differs from ϵ_F . Comment on the numerical significance of this "failure" of the Sommerfeld expansion, and on the mathematical reason for the "failure."

2. Thermodynamics of the Free and Independent Electron Gas

- Deduce from the thermodynamic identities

$$c_v = \left(\frac{\partial u}{\partial T} \right)_n = T \left(\frac{\partial s}{\partial T} \right)_n, \quad (2.96)$$

³¹ The Lorentz model is, however, of considerable importance in the description of semiconductors (Chapter 29).

³² The thermopower is a notable exception.

³³ These assertions are correct to leading order in $k_B T/\epsilon_F$, but in metals this is always a good expansion parameter.

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from Eqs. (2.56) and (2.57), and from the third law of thermodynamics ($s \rightarrow 0$ as $T \rightarrow 0$) that the entropy density, $s = S/V$ is given by:

$$s = -k_B \int \frac{d\mathbf{k}}{4\pi^3} [f \ln f + (1-f) \ln (1-f)], \quad (2.97)$$

where $f(\epsilon(\mathbf{k}))$ is the Fermi function (Eq. (2.56)).

(b) Since the pressure P satisfies Eq. (B.5) in Appendix B, $P = -(u - Ts - \mu n)$, deduce from (2.97) that

$$P = k_B T \int \frac{d\mathbf{k}}{4\pi^3} \ln \left(1 + \exp \left[-\frac{(\hbar^2 k^2/2m) - \mu}{k_B T} \right] \right). \quad (2.98)$$

Show that (2.98) implies that P is a homogeneous function of μ and T of degree $5/2$; that is,

$$P(\lambda\mu, \lambda T) = \lambda^{5/2} P(\mu, T) \quad (2.99)$$

for any constant λ .

(c) Deduce from the thermodynamic relations in Appendix B that

$$\left(\frac{\partial P}{\partial \mu} \right)_T = n, \quad \left(\frac{\partial P}{\partial T} \right)_\mu = s. \quad (2.100)$$

(d) By differentiating (2.99) with respect to λ show that the ground-state relation (2.34) holds at any temperature, in the form

$$P = \frac{2}{3} u. \quad (2.101)$$

(e) Show that when $k_B T \ll \epsilon_F$, the ratio of the constant-pressure to constant-volume specific heats satisfies

$$\left(\frac{c_p}{c_v} \right) - 1 = \frac{\pi^2}{3} \left(\frac{k_B T}{\epsilon_F} \right)^2 + O \left(\frac{k_B T}{\epsilon_F} \right)^4.$$

(f) Show, by retaining further terms in the Sommerfeld expansions of u and n , that correct to order T^3 the electronic heat capacity is given by

$$c_v = \frac{\pi^2}{3} k_B^2 T g(\epsilon_F) - \frac{\pi^4}{90} k_B^4 T^3 g(\epsilon_F) \left[15 \left(\frac{g'(\epsilon_F)}{g(\epsilon_F)} \right)^2 - 21 \frac{g''(\epsilon_F)}{g(\epsilon_F)} \right]. \quad (2.102)$$

3. The Classical Limit of Fermi-Dirac Statistics

The Fermi-Dirac distribution reduces to the Maxwell-Boltzmann distribution, provided that the Fermi function (2.56) is much less than unity for every positive ϵ , for in that case we must have

$$f(\epsilon) \approx e^{-(\epsilon - \mu)/k_B T}. \quad (2.103)$$

The necessary and sufficient condition for (2.103) to hold for all positive ϵ is

$$e^{-\mu/k_B T} \gg 1. \quad (2.104)$$

(a) Assuming that (2.104) holds, show that

$$r_s = e^{-\mu/3k_B T} 3^{1/3} \pi^{1/6} \hbar (2mk_B T)^{-1/2}. \quad (2.105)$$

In conjunction with (2.104) this requires that

$$r_s \gg \left(\frac{\hbar^2}{2mk_B T} \right)^{1/2}, \quad (2.106)$$

which can also be taken as the condition for the validity of classical statistics.

(b) What is the significance of the length r_s must exceed?

(c) Show that (2.106) leads to the numerical condition

$$\frac{r_s}{a_0} \gg \left(\frac{10^5 \text{ K}}{T} \right)^{1/2}. \quad (2.107)$$

(d) Show that the normalization constant $m^3/4\pi^3\hbar^3$ appearing in the Fermi-Dirac velocity distribution (2.2) can also be written as $(3\sqrt{\pi}/4)n(m/2\pi k_B T)^{3/2}$ so that $f_B(0)/f(0) = (4/3\sqrt{\pi})(T_F/T)^{3/2}$.

4. Insensitivity of the Distribution Function to Small Changes in the Total Number of Electrons

In deriving the Fermi distribution (page 41) we argued that the probability of a given level being occupied should not change appreciably when the total number of electrons is changed by one. Verify that the Fermi function (2.56) is compatible with this assumption as follows:

(a) Show, when $k_B T \ll \epsilon_F$, that when the number of electrons changes by one at fixed temperature, the chemical potential changes by

$$\Delta\mu = \frac{1}{Vg(\epsilon_F)}, \quad (2.108)$$

where $g(\epsilon)$ is the density of levels.

(b) Show, as a consequence of this, that the most the probability of any level being occupied can change by is

$$\Delta f = \frac{1}{6} \frac{\epsilon_F}{k_B T} \frac{1}{N}. \quad (2.109)$$

[Use the free electron evaluation (2.65) of $g(\epsilon_F)$.] Although temperatures of millidegrees Kelvin can be reached, at which $\epsilon_F/k_B T \approx 10^8$, when N is of order 10^{22} then Δf is still negligibly small.

Appendix C

The Sommerfeld Expansion

The Sommerfeld expansion is applied to integrals of the form

$$\int_{-\infty}^{\infty} d\varepsilon H(\varepsilon) f(\varepsilon), \quad f(\varepsilon) = \frac{1}{e^{(\varepsilon - \mu)/k_B T} + 1}, \quad (\text{C.1})$$

where $H(\varepsilon)$ vanishes as $\varepsilon \rightarrow -\infty$ and diverges no more rapidly than some power of ε as $\varepsilon \rightarrow +\infty$. If one defines

$$K(\varepsilon) = \int_{-\infty}^{\varepsilon} H(\varepsilon') d\varepsilon', \quad (\text{C.2})$$

so that

$$H(\varepsilon) = \frac{dK(\varepsilon)}{d\varepsilon}, \quad (\text{C.3})$$

then one can integrate by parts¹ in (C.1) to get

$$\int_{-\infty}^{\infty} H(\varepsilon) f(\varepsilon) d\varepsilon = \int_{-\infty}^{\infty} K(\varepsilon) \left(-\frac{\partial f}{\partial \varepsilon} \right) d\varepsilon. \quad (\text{C.4})$$

Since f is indistinguishable from zero when ε is more than a few $k_B T$ greater than μ , and indistinguishable from unity when ε is more than a few $k_B T$ less than μ , its ε -derivative will be appreciable only within a few $k_B T$ of μ . Provided that H is nonsingular and not too rapidly varying in the neighborhood of $\varepsilon = \mu$, it is very reasonable to evaluate (C.4) by expanding $K(\varepsilon)$ in a Taylor series about $\varepsilon = \mu$, with the expectation that only the first few terms will be of importance:

$$K(\varepsilon) = K(\mu) + \sum_{n=1}^{\infty} \left[\frac{(\varepsilon - \mu)^n}{n!} \right] \left[\frac{d^n K(\varepsilon)}{d\varepsilon^n} \right]_{\varepsilon=\mu}. \quad (\text{C.5})$$

When we substitute (C.5) in (C.4), the leading term gives just $K(\mu)$, since

$$\int_{-\infty}^{\infty} (-\partial f / \partial \varepsilon) d\varepsilon = 1.$$

Furthermore, since $\partial f / \partial \varepsilon$ is an even function of $\varepsilon - \mu$, only terms with even n in (C.5) contribute to (C.4), and if we reexpress K in terms of the original function H through (C.2), we find that:

$$\begin{aligned} \int_{-\infty}^{\infty} d\varepsilon H(\varepsilon) f(\varepsilon) &= \int_{-\infty}^{\mu} H(\varepsilon) d\varepsilon \\ &+ \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \frac{(\varepsilon - \mu)^{2n}}{(2n)!} \left(-\frac{\partial f}{\partial \varepsilon} \right) d\varepsilon \frac{d^{2n-1}}{d\varepsilon^{2n-1}} H(\varepsilon) \Big|_{\varepsilon=\mu}. \end{aligned} \quad (\text{C.6})$$

¹ The integrated term vanishes at ∞ because the Fermi function vanishes more rapidly than K diverges, and at $-\infty$ because the Fermi function approaches unity while K approaches zero.

Finally, making the substitution $(\varepsilon - \mu)/k_B T = x$, we find that

$$\int_{-\infty}^{\infty} H(\varepsilon) f(\varepsilon) d\varepsilon = \int_{-\infty}^{\mu} H(\varepsilon) d\varepsilon + \sum_{n=1}^{\infty} a_n (k_B T)^{2n} \frac{d^{2n-1}}{d\varepsilon^{2n-1}} H(\varepsilon) \Big|_{\varepsilon=\mu}, \quad (\text{C.7})$$

where the a_n are dimensionless numbers given by

$$a_n = \int_{-\infty}^{\infty} \frac{x^{2n}}{(2n)!} \left(-\frac{d}{dx} \frac{1}{e^x + 1} \right) dx. \quad (\text{C.8})$$

By elementary manipulations one can show that

$$a_n = 2 \left(1 - \frac{1}{2^{2n}} + \frac{1}{3^{2n}} - \frac{1}{4^{2n}} + \frac{1}{5^{2n}} - \cdots \right). \quad (\text{C.9})$$

This is usually written in terms of the Riemann zeta function, $\zeta(n)$, as

$$a_n = \left(2 - \frac{1}{2^{2(n-1)}} \right) \zeta(2n), \quad (\text{C.10})$$

where

$$\zeta(n) = 1 + \frac{1}{2^n} + \frac{1}{3^n} + \frac{1}{4^n} + \cdots. \quad (\text{C.11})$$

For the first few n , $\zeta(2n)$ has the values²

$$\zeta(2n) = 2^{2n-1} \frac{\pi^{2n}}{(2n)!} B_n \quad (\text{C.12})$$

where the B_n are known as Bernoulli numbers, and

$$B_1 = \frac{1}{6}, \quad B_2 = \frac{1}{30}, \quad B_3 = \frac{1}{42}, \quad B_4 = \frac{1}{30}, \quad B_5 = \frac{5}{66}. \quad (\text{C.13})$$

In most practical calculations in metals physics, one rarely needs to know more than $\zeta(2) = \pi^2/6$, and never goes beyond $\zeta(4) = \pi^4/90$. Nevertheless, if one should wish to carry the Sommerfeld expansion (2.70) beyond $n = 5$ (and hence past the values of the B_n listed in (C.13)), by the time $2n$ is as large as 12 the a_n can be evaluated to five-place accuracy by retaining only the first two terms in the alternating series (C.9).

² See, for example, E. Jahnke and F. Emde, *Tables of Functions*, 4th ed., Dover, New York, 1945, p. 272.